



Mahnoor Rashad

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Gender: Female **Date of birth:** 20/04/1998 **Place of birth:** Lahore, Pakistan

Nationality: Pakistani

ABOUT ME

Physics researcher with expertise in nanomaterials, electrochemical energy storage, ferrite materials, and advanced electrode materials for supercapacitors. My research focuses on the synthesis, characterization, and performance optimization of metal oxide heterostructures and nanocomposites for sustainable energy storage, with hands-on experience in the co-precipitation synthesis method and ion doping strategies. I am interested in developing next-generation materials for supercapacitors, batteries, and other electrochemical energy devices. MPhil and BS in Physics from The University of Lahore, with a first-author publication and a second manuscript in preparation. Seeking a PhD position in Physics / Materials Science to build on this research record.

SKILLS

Nanomaterial synthesis (co-precipitation method) | Metal oxide heterostructures & nanocomposites | Ion doping & structural modification | Supercapacitor / electrochemical characterization | Materials chemistry & electrochemistry | Data analysis & scientific writing | Physics laboratory techniques | Critical thinking & problem-based research | Microsoft Office (Word, Excel, PowerPoint) | Presentation & academic communication skills

WORK EXPERIENCE

Graduate Researcher

City: Lahore | **Country:** Pakistan

[25/02/2024 - Current]

Mphil thesis project, The University of Lahore

Synthesized pure $\text{NiSnFe}_2\text{O}_4$ nanoparticles using the co-precipitation method for supercapacitor electrode applications.

Investigated the effect of Bismuth (Bi) ion doping at varying concentrations on capacitance and energy-storage performance.

Carried out structural and electrochemical characterization to evaluate and optimize supercapacitor performance.

Findings contributed to a peer-reviewed publication and a second manuscript currently in preparation.

EDUCATION & TRAINING

[17/02/2024 - 17/12/2025]

MPhil Physics

The University of Lahore

City: Lahore | **Country:** Pakistan | **Final grade:** 3.88/4.0 | **Level in EQF:** 7

[06/09/2021 - 10/01/2024]

BS Physics

The University of Lahore

City: Lahore | **Country:** Pakistan | **Final grade:** 3.72/4.0 | **Level in EQF:** 6

[02/04/2025 - Current]

Bachelor of Education (B.Ed.) Secondary

Virtual university

City: Lahore | Country: Pakistan | Level in EQF: 6

LANGUAGE SKILLS

Mother tongue(s): Urdu

Other language(s):

English

LISTENING: C1 **READING:** C1 **WRITING:** B2

SPOKEN PRODUCTION: B2 **SPOKEN INTERACTION:** B2

PUBLICATIONS

[2026] [Engineering Fe₂WO₆-based heterostructures for high-performance supercapacitors: the role of V₂O₅ and g-C₃N₄ integration](#)

Reference: Mahnoor Rashad, 2026, Current Status: Published

Journal Name: Physical Chemistry Chemical Physics (Royal Society of Chemistry)

[2026] **Modified Structure of Nickel Tin Ferrite by Bismuth Doping for High-Performance Supercapacitor**

Reference: Mahnoor Rashad, 2026, Current Status: In process for publish

CONFERENCES & SEMINARS

[08/10/2026 - 09/10/2026] **International Conference on Emerging Trends in Physics (ICETP) 2024** The University of Lahore

PROJECTS

[20/02/2025 - 04/01/2026] **Structural modification of nickel tin ferrites via bismuth doping for supercapacitors**

As part of my MPhil research, I focused on the synthesis of (NiSnFe₂O₄) nanoparticles improved energy storage for supercapacitor applications . Using the Co-precipitation method , synthesizing pure NiSnFe₂O₄ nanoparticles and investigating the impact of doping them with Bismuth (Bi) ions at various concentrations to enhance their capacitance and storage capacity.

[01/11/2023 - 01/01/2024] **Anti-Sleep-Glasses**

As part of my undergraduate final year project, I developed a wearable anti-sleep alarm system designed to detect early signs of driver drowsiness and alert them in real-time. The device aimed to reduce the risk of accidents caused by dozy driving, enhancing road safety for long-distance and night-time drivers.

HONOURS AND AWARDS

[15/05/2026] **Bronze Medal – MPhil Physics Awarding institution:** The University of Lahore

Awarded the Bronze Medal for achieving **third position** in the MPhil Physics program through merit-based evaluation of academic performance and research accomplishments at The University of Lahore.

RECOMMENDATIONS

Name: Prof. Dr. Muhammad Iftikhar Khan Supervisor of Mphil research

MPhil research supervisor who supervised my research in Physics and can attest to my research skills, academic performance, laboratory experience, and professional work ethic.

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